Data Science

CSE558

Recap

- Chi Square Test

- Mean & Proportion Test for pair of sample sets.

Overview

- Hypothesis Testing

- Probability

t-test for Correlation Coefficient

- Investigate if sample correlation coefficient indicates dependence.
- Standard deviation of populations are unknown.

From two population of n independent samples $\{x_i, y_i\}$. Let r be the correlation coefficient.

H_0: $r = 0$; H_1: $r \neq 0_2$;

Level of significance: $\propto = 0.05$

Limitations: Uncovers only linear relationship.

Let $r = \dfrac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\left[\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2\right]^{\frac{1}{2}}}$

Then, the test statistics $t = \dfrac{r}{\sqrt{1 - r^2}} \cdot \sqrt{n - 2}$. Degrees of freedom = n-2

Example

A sample of 15 men 50 years and older was taken and the average number of cigarettes smoked per day and the age at death was recorded. With a 5% level of significance what can you conclude?

Cigarettes	5	23	25	48	17	8	4	26	11	19	14	35	29	4	23
Longevity	80	78	60	53	85	84	73	79	81	75	68	72	58	92	65

What does one-sided t-test for correlation coefficient mean?

z-test for Two Counts (Poisson)

- Investigates if difference between two counts are significant.

Let n_1 and n_2 be two outcomes over time t_1 and t_2 respectively.
H_0: $\mu_1 = \mu_2$; H_1: $\mu_1 \neq \mu_2$;
Level of significance: $\propto = 0.05$

Let, $R_1 = n_1/t_1$ and $R_2 = n_2/t_2$, then the test statistics is $Z = \dfrac{(R_1 - R_2)}{\left(\dfrac{R_1}{t_1} + \dfrac{R_2}{t_2}\right)^{\frac{1}{2}}}$

F-test for Two Population Variance

- Investigate the difference between two population variances.
- Use when populations are assumed from normal distribution.

From two populations, let $x_1, x_2, \ldots, x_{n1}$ and $y_1, y_2, \ldots, y_{n2}$ are sampled independently.

H_0: $\sigma_1^2 = \sigma_2^2$; H_1: $\sigma_1^2 \neq \sigma_2^2$;

Level of significance: $\propto = 0.05$

Let $\bar{x} = \dfrac{\sum x_i}{n_1}, \quad \bar{y} = \dfrac{\sum y_i}{n_2}$ and $s_1^2 = \dfrac{\sum (x_i - \bar{x})^2}{n_1 - 1}, \quad s_2^2 = \dfrac{\sum (y_i - \bar{y})^2}{n_2 - 1}$

Then, the test statistics $F = \dfrac{s_1^2}{s_2^2}$. Degrees of freedom = $(n_1 - 1, n_2 - 1)$

Means need not be same.

χ^2-test for Goodness of Fit

- Investigate the significance of the difference between observed frequency and expected frequencies in K classes.
- Used when expected frequency is high (> 5; in practice).

H_0: population follows a distribution; H_1: does not follow;

Level of significance: $\propto$ = 0.05

The test statistics is. $\chi^2 = \sum_{i=1}^{K} \frac{(O_i - E_i)^2}{E_i}$

- Degrees of freedom = K-1
- If, observed frequencies are sample of m parameters then, degrees of freedom = K - 1 - m

Relation to Normal Distribution

$$Z = \frac{np - np_0}{\sqrt{np_0(1 - p_0)}} \qquad Z^2 = \frac{(np - np_0)^2}{np_0(1 - p_0)}$$

Similarly compute Z-statistic for the complement event.

Their sum is, $\dfrac{(np - np_0)^2}{np_0} + \dfrac{(n - np - n(1 - p_0))^2}{n(1 - p_0)}$

$$\chi^2 = \sum_{i=1}^{K} \frac{(O_i - E_i)^2}{E_i}$$

χ^2-test for Goodness of Fit

There is a theory that about 10% people are left handed in the world. In your class of 140 students, you found 21 are left handed. Perform test to check if the class fits the theory with 5% level of significance.

	Observed	Expected
Left	21	14
Right	119	126

	Level of significance α							
Two-sided One-sided	0.20 0.10		0.10 0.05		0.05 0.025		0.01 0.005	
ν	a	b	a	b	a	b	a	b
1	0.016	2.71	39.10^{-4}	3.84	98.10^{-5}	5.02	16.10^{-5}	6.63
2	0.21	4.61	0.10	5.99	0.05	7.38	0.02	9.21
3	0.58	6.25	0.35	7.81	0.22	9.35	0.11	11.34
4	1.06	7.78	0.71	9.49	0.48	11.14	0.30	13.28
5	1.61	9.24	1.15	11.07	0.83	12.83	0.55	15.09

Example

A random sample of n = 60 printed circuit boards is taken and the number of defects recorded. Test, with a 5% level of significance if the observation follows poisson distribution.

#Defects	0	1	2	3
#Frequency	32	15	9	4

Hypothesis?

Compute expected values.

χ^2-test for Goodness of Fit

H_0: $p_1 = 0.1, p_2 = 0.3, p_3 = 0.6$; H_1: Does not follow this distribution;

When to use one sided test?

H_0: $p_1 < 0.1, p_2 < 0.3, p_3 < 0.6$; H_1: Does not follow this distribution;

Recall Independence

If two events A and B are independent, then $P(A \cap B) = P(A) \cdot P(B)$

Say you win a lottery, if you get 6 rolling an unbiased die and get head on tossing an unbiased coin.

$P(6 \cap \text{head}) = \frac{1}{6} \cdot \frac{1}{2} = 1/12$. (Since rolling die is independent of tossing coin)

Now, if you repeat this event 48 times, then over expectation you win the lottery 4 times.

If events are independent then

$E(A \cap B) = P(A) \cdot P(B) \cdot \text{Count}$

χ^2-test for Independence

From a survey a 120 people for their preferred social media platform. Is there enough evidence (5% level of significance) to suggest that social media preference is independent of sex?

Observed	Male	Female	Total
Tik Tok	15	20	35
Instagram	30	35	65
Facebook	5	15	20
Total	50	70	120

Expected	Male	Female	Total
Tik Tok	14.6	20.4	35
Instagram	27.1	37.9	65
Facebook	8.3	11.7	20
Total	50	70	120

E(TikTok and Male) = $50/120 \cdot 35/120 \cdot 120 = 14.6$

$\chi^2 = 2.84$. Degrees of freedom = $(r-1) \cdot (c-1)$

Overview

- Hypothesis Testing

- Probability Theory

Bounding Probabilities of Events

Let $(\Omega, \mathcal{F}, \mathbf{P})$ be a probability space. Let $\mathbf{A}_1, \mathbf{A}_2, \ldots, \mathbf{A}_n \subseteq \mathcal{F}$. Then

1. Upper bound: $\mathbf{P}\left(\bigcup_{i=1}^{n} \mathbf{A}_i\right) \leq \sum_{i=1}^{n} \mathbf{P}(\mathbf{A}_i)$

Proof: By Induction

n = 1 (trivial)

n = 2; $\mathbf{P}(\mathbf{A} \cup \mathbf{B}) = \mathbf{P}(\mathbf{A}) + \mathbf{P}(\mathbf{B}) - \mathbf{P}(\mathbf{A} \cap \mathbf{B}) \leq \mathbf{P}(\mathbf{A}) + \mathbf{P}(\mathbf{B})$

Bounding Probabilities of Events

Let $(\Omega, \mathcal{F}, \mathbf{P})$ be a probability space. Let $\mathbf{A}_1, \mathbf{A}_2, \ldots, \mathbf{A}_n \subseteq \mathcal{F}$. Then

1. Upper bound: $$\mathbf{P}\left(\bigcup_{i=1}^{n} \mathbf{A}_i\right) \leq \sum_{i=1}^{n} \mathbf{P}(\mathbf{A}_i)$$

Proof: By Induction

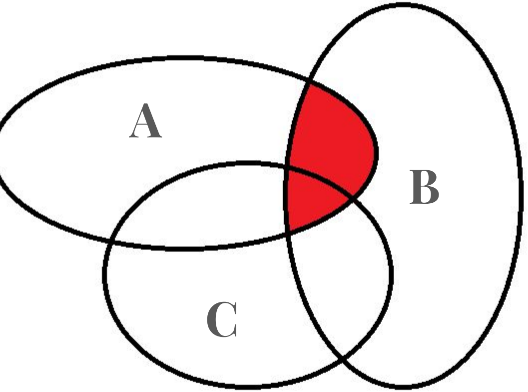
n = 2; $\mathbf{P}(\mathbf{A} \cup \mathbf{B}) \leq \mathbf{P}(\mathbf{A}) + \mathbf{P}(\mathbf{B})$

For n = k let, $$\mathbf{P}\left(\bigcup_{i=1}^{k} \mathbf{A}_i\right) \leq \sum_{i=1}^{k} \mathbf{P}(\mathbf{A}_i)$$

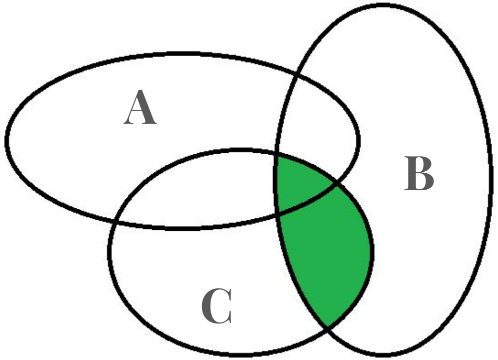
So, for n = k + 1,

$$\mathbf{P}\left(\bigcup_{i=1}^{k+1} \mathbf{A}_i\right) = \mathbf{P}\left(\left\{\bigcup_{i=1}^{k} \mathbf{A}_i\right\} \bigcup \mathbf{A}_{k+1}\right) \leq \mathbf{P}\left(\left\{\bigcup_{i=1}^{k} \mathbf{A}_i\right\}\right) + \mathbf{P}(\mathbf{A}_{k+1}) \leq \sum_{i=1}^{k+1} \mathbf{P}(\mathbf{A}_i)$$

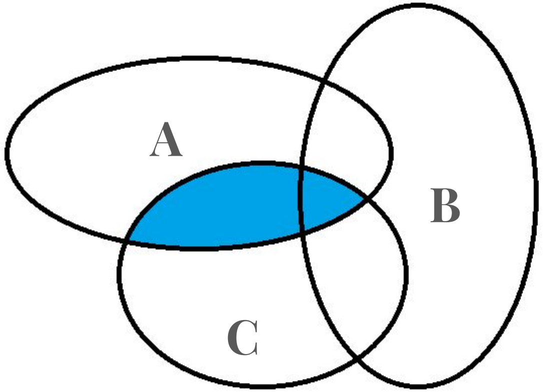
Principle of Inclusion & Exclusion



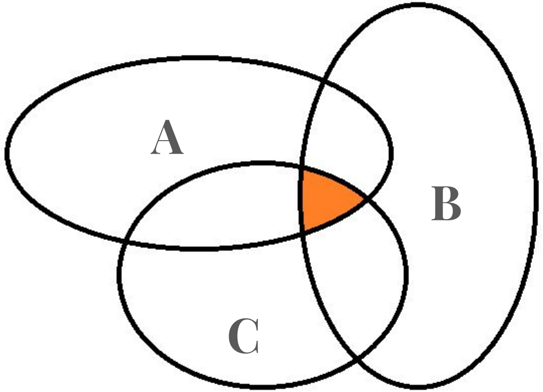
$A \cap B$



$B \cap C$

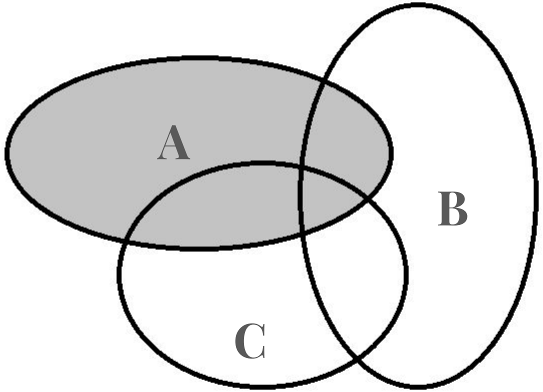


$C \cap A$



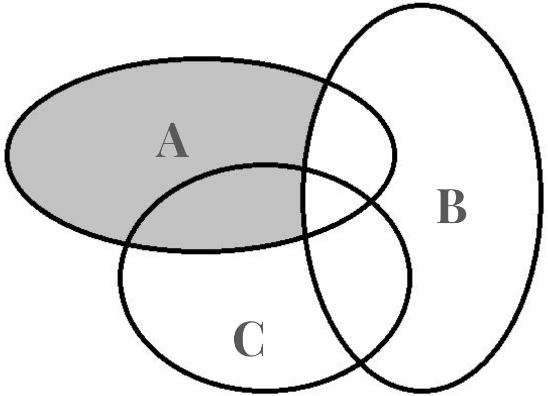
$A \cap B \cap C$

Principle of Inclusion & Exclusion



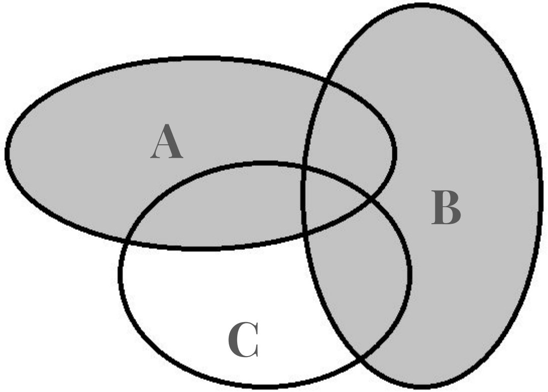
$P(A)$

Principle of Inclusion & Exclusion



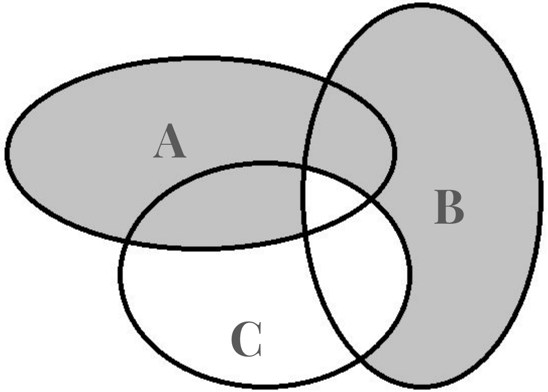
$P(A) - P(A \cap B)$

Principle of Inclusion & Exclusion



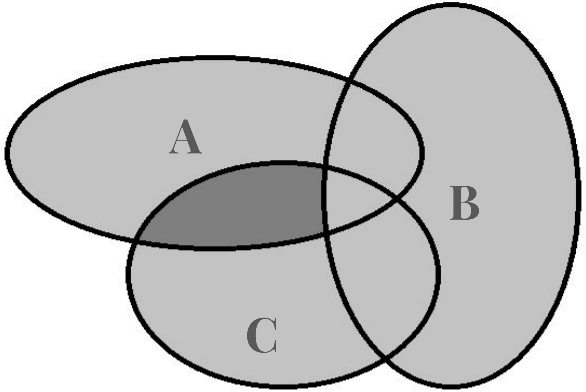
$P(A) - P(A \cap B) + P(B)$

Principle of Inclusion & Exclusion



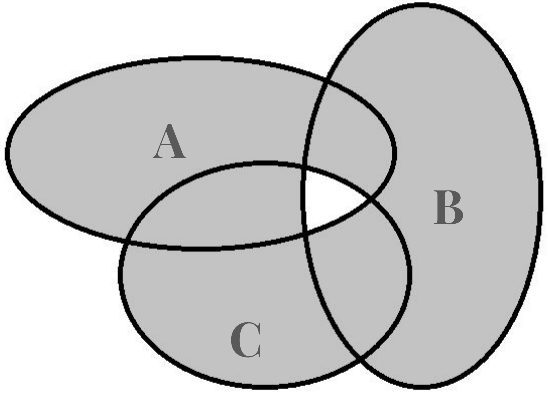
$P(A) - P(A \cap B) + P(B) - P(B \cap C)$

Principle of Inclusion & Exclusion



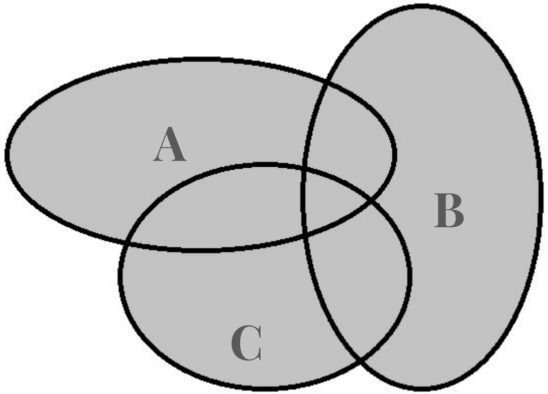
$P(A) - P(A \cap B) + P(B) - P(B \cap C) + P(C)$

Principle of Inclusion & Exclusion



$P(A) - P(A \cap B) + P(B) - P(B \cap C) + P(C) - P(C \cap A)$

Principle of Inclusion & Exclusion



$P(A) - P(A \cap B) + P(B) - P(B \cap C) + P(C) - P(C \cap A) + P(A \cap B \cap C)$

$P(A \cup B \cup C) = P(A) + P(B) + P(C) - (P(A \cap B) + P(B \cap C) + P(C \cap A)) + P(A \cap B \cap C)$

Principle of Inclusion & Exclusion

$$\left|\bigcup_{i=1}^{n} A_i\right| = \sum_{i=1}^{n} |A_i| - \sum_{1 \leqslant i < j \leqslant n} |A_i \cap A_j| + \sum_{1 \leqslant i < j < k \leqslant n} |A_i \cap A_j \cap A_k| - \cdots + (-1)^{n+1} |A_1 \cap \cdots \cap A_n|$$

$$\left|\bigcup_{i=1}^{n} A_i\right| = \sum_{k=1}^{n} (-1)^{k+1} \left(\sum_{1 \leqslant i_1 < \cdots < i_k \leqslant n} |A_{i_1} \cap \cdots \cap A_{i_k}| \right)$$

Bounding Probabilities of Events

Let $(\Omega, \mathcal{F}, \mathbf{P})$ be a probability space. Let $A_1, A_2, \ldots, A_n \subseteq \mathcal{F}$. Then

1. Upper bound: $$\mathbf{P}\left(\bigcup_{i=1}^{n} \mathbf{A}_i\right) \leq \sum_{i=1}^{n} \mathbf{P}(\mathbf{A}_i)$$
2. Lower bound: $$\mathbf{P}\left(\bigcup_{i=1}^{n} \mathbf{A}_i\right) \geq \sum_{i=1}^{n} \mathbf{P}(\mathbf{A}_i) - \sum_{1 \leq i,j \leq n} \mathbf{P}(\mathbf{A}_i \cap \mathbf{A}_j)$$

Proof: By induction

n = 1 & 2 (trival)

n = 3 (Principle of inclusion & exclusion)

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - (P(A \cap B) + P(B \cap C) + P(C \cap A)) + P(A \cap B \cap C)$$
$$\geq P(A) + P(B) + P(C) - (P(A \cap B) + P(B \cap C) + P(C \cap A))$$

Bounding Probabilities of Events

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $A_1, A_2, \ldots, A_n \subseteq \mathcal{F}$. Then

1. Upper bound: $$\mathbf{P}\left(\bigcup_{i=1}^{n} \mathbf{A}_i\right) \leq \sum_{i=1}^{n} \mathbf{P}(\mathbf{A}_i)$$
2. Lower bound: $$\mathbf{P}\left(\bigcup_{i=1}^{n} \mathbf{A}_i\right) \geq \sum_{i=1}^{n} \mathbf{P}(\mathbf{A}_i) - \sum_{1 \leq i,j \leq n} \mathbf{P}(\mathbf{A}_i \cap \mathbf{A}_j)$$

Proof: By induction

n = 1 & 2 (trival)

n = 3; $P(A \cup B \cup C) \geq P(A) + P(B) + P(C) - (P(A \cap B) + P(B \cap C) + P(C \cap A))$

Let, for n = k, $$\mathbf{P}\left(\bigcup_{i=1}^{k} \mathbf{A}_i\right) \geq \sum_{i=1}^{k} \mathbf{P}(\mathbf{A}_i) - \sum_{1 \leq i,j \leq k} \mathbf{P}(\mathbf{A}_i \cap \mathbf{A}_j)$$

Bounding Probabilities of Events

$$\mathbf{P}\left(\bigcup_{i=1}^{n} \mathbf{A}_i\right) \leq \sum_{i=1}^{n} \mathbf{P}(\mathbf{A}_i)$$

Proof: By induction

For n = k + 1

$$\begin{aligned}
\mathbf{P}\left(\bigcup_{i=1}^{k+1} \mathbf{A}_i\right) &= \mathbf{P}\left(\left\{\bigcup_{i=1}^{k} \mathbf{A}_i\right\}\right) + \mathbf{P}(\mathbf{A}_{k+1}) - \mathbf{P}\left(\bigcup_{i=1}^{k} (\mathbf{A}_i \cap \mathbf{A}_{k+1})\right) \\
&\geq \sum_{i=1}^{k+1} \mathbf{P}(\mathbf{A}_i) - \sum_{1 \leq i,j \leq k} \mathbf{P}(\mathbf{A}_i \cap \mathbf{A}_j) - \mathbf{P}\left(\bigcup_{i=1}^{k} (\mathbf{A}_i \cap \mathbf{A}_{k+1})\right) \\
&\geq \sum_{i=1}^{k+1} \mathbf{P}(\mathbf{A}_i) - \sum_{1 \leq i,j \leq k} \mathbf{P}(\mathbf{A}_i \cap \mathbf{A}_j) - \sum_{i=1}^{k} \mathbf{P}\left(\mathbf{A}_i \cap \mathbf{A}_{k+1}\right) \\
&= \sum_{i=1}^{k+1} \mathbf{P}(\mathbf{A}_i) - \sum_{1 \leq i,j \leq k+1} \mathbf{P}(\mathbf{A}_i \cap \mathbf{A}_j)
\end{aligned}$$

Bounding Probabilities of Events

Let $(\Omega, \mathcal{F}, \mathbf{P})$ be a probability space. Let $\mathbf{A}_1, \mathbf{A}_2, \ldots, \mathbf{A}_n \subseteq \mathcal{F}$. Then

1. Upper bound: $$\mathbf{P}\left(\bigcup_{i=1}^{n} \mathbf{A}_i\right) \leq \sum_{i=1}^{n} \mathbf{P}(\mathbf{A}_i)$$
2. Lower bound: $$\mathbf{P}\left(\bigcup_{i=1}^{n} \mathbf{A}_i\right) \geq \sum_{i=1}^{n} \mathbf{P}(\mathbf{A}_i) - \sum_{1 \leq i,j \leq n} \mathbf{P}(\mathbf{A}_i \cap \mathbf{A}_j)$$